

USI, Risk Management, Spring 2025

Case #2

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Due by 23:59 on May 21, 2025

You are the risk manager of a Swiss-based hedge fund company managing an equity and a fixed-income fund. It's closing of **April 25, 2025** and you need to compute risk measures on your portfolio.

Part I: Equity Option Fund

The option portfolio is reported in spreadsheet "Portfolio" of **Data_Case2.xlsx**, and summarized below:

- long 7'000 put options on *Nestle* (NESN)
- short 2'000 call options on *Swiss Re* (SREN).

Each contract allows you to take position on 1'000 units of the underlying. Further details on the options are in spreadsheet "Option_Portfolio" of **Data_Case2.xlsx**, and information about the underlying stocks are in spreadsheet "Underlying_Assets" of **Data_Case2.xlsx**.

1. Calculate the option portfolio $\text{VaR}(99\%,1)$ and $\text{ES}(99\%,1)$ using an Asset-Normal approach, where you map the options to each underlying stock.
2. Now, you assume that the main source of risk in the equity option portfolio is given by exposure to the Swiss Market Index (SMI). Given underlying stocks' β s in spreadsheet "Underlying_Assets" of **Data_Case2.xlsx**, use the mapping (to the underlying, and then to SMI) approach to obtain the $\text{VaR}(99\%,1)$ and $\text{ES}(99\%,1)$ of the option portfolio.
3. As an alternative, you rely on a Historical Simulation Approach where you construct 250 hypothetical next day' prices to the two stocks based on (joint) stock returns in the last 250 days, keeping the risk-free rate and implied volatility unchanged. Compute the simulated value of the portfolio in correspondence to each simulation. From that, obtain VaR and ES of the portfolio. Compare to those previously obtained.

Part II: Fixed Income Fund

The USD-denominated fixed-income positions as of April 25, 2025 (closing) consist of US government bonds, and are reported in spreadsheet “FI_Portfolio ” of **Data_Case2.xlsx**.

You want to evaluate the riskiness of the portfolio through the 1-day VaR and expected shortfall at the 99% confidence level. Spreadsheet “TSIR” contains data on the US term structure of yields (i.e., zcb spot rates, in percentage, discrete compounding). Based on this information:

1. Compute the individual bond prices as present discounted values of their cash-flows using today’s term structure, and hence the overall portfolio value.
2. Use a cash-flow mapping approach where you map all positions to the relevant maturities to compute the Asset Normal VaR and ES. To this end, use the sample VCV matrix computed on past 250-day yield *changes*.
3. Next, you rely on a Historical Simulation Approach where you construct 250 hypothetical next day’ term structures based on (joint) yield *changes* in the last 250 days. Compute the simulated value of the portfolio in correspondence to each simulated term structure. From that, obtain VaR and ES. Compare to the VaR and ES previously obtained. For simplicity, ignore the passage of one day in the maturity of the bonds when recomputing the portfolio value.

Part III: Total Fund risk

For regulatory reporting purpose, you must provide the $\text{VaR}(99\%,1)$ and $\text{ES}(99\%,1)$ for the entire fund composed of the equity plus fixed-income portfolios. Starting from the risk measures you computed in point 3 of each part, obtain the aggregate figures. State any assumption you deem necessary and motivate your answer and computations. In doing this, you shall explicitly or implicitly account for the fact that your reporting is in CHF while part of the portfolio is USD-denominated.